



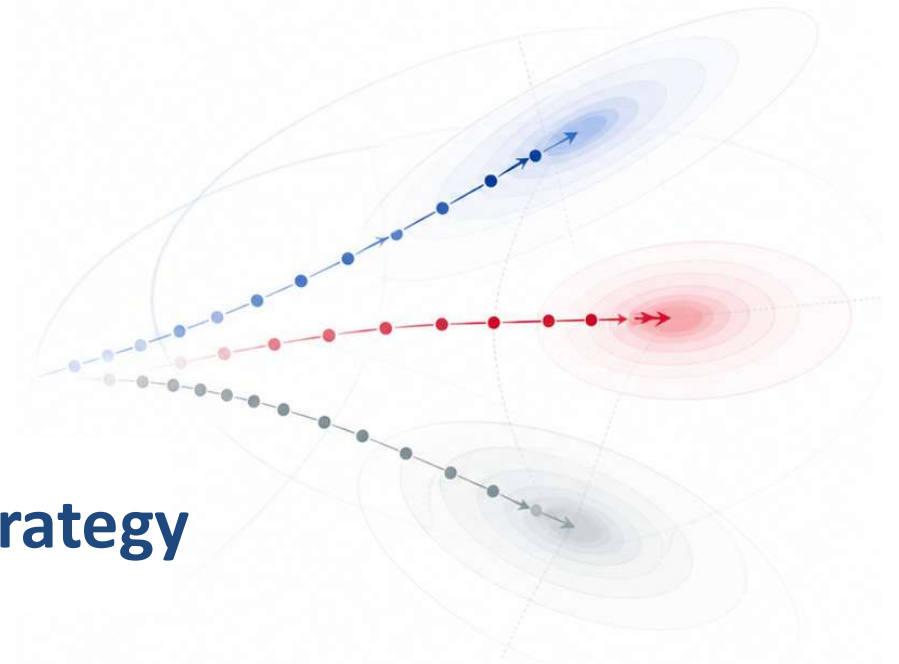
Genetic and Evolutionary  
Computation Conference



San José, Costa Rica  
July 13-17, 2026

# S3-LRA-xNES

Step-size and Shape Separated  
Learning Rate Adaptation for  
Exponential Natural Evolution Strategy



*Kosuke Ujihara, Masahiro Nomura, Isao Ono*

*Institute of Science Tokyo*

*Yokohama, Japan*

[ujihara.k@ic.comp.isct.ac.jp](mailto:ujihara.k@ic.comp.isct.ac.jp)

# One-Minute Summary:

## Core Idea and Main Results

---

**Core Idea:** To apply a well-known principle from ESs  
–separating dynamics of step-size and shape–  
to learning rate adaptation.

**Main Results:** Comparison with LRA-CMA-ES [Nomura 24]

ILL-CONDITIONED

**82.4%**

Median reduction  
in avg. evaluations

MULTIMODAL

**50/50**

successes on  
80-D Schaffer

**LRA-CMA-ES: 0/50**

KEY DESIGN CHANGE

**Separating  
adaptation of  
learning rates**

**$\eta_\sigma$  for step-size**

**$\eta_B$  for shape**

## One-Minute Summary: Core Idea and Main Results

---

**Core Idea:** To apply a well-known principle from ESs  
–separating dynamics of step-size and shape–  
to learning rate adaptation.

Main Results: Comparison with LRA-CMA-ES [Nomura 24]

ILL-CONDITIONED

**82.4%**

Median reduction  
in avg. evaluations

MULTIMODAL

**50/50**

successes on  
80-D Schaffer

**LRA-CMA-ES: 0/50**

KEY DESIGN CHANGE

**Separating  
adaptation of  
learning rates**

**$\eta_\sigma$  for step-size**

**$\eta_B$  for shape**

Speed and robustness improve for the same reason:  
step-size and shape can react independently.

# Problem in LRA-CMA-ES: Shared $\eta_\Sigma$ Breaks the Separation Principle

---

## STANDARD CMA-ES / xNES

**Step-size**

global scale

$\eta_\sigma$

**Shape**

anisotropy

$\eta_B$

**step-size & shape updates:  
deliberately decoupled**

$\neq$

## LRA-CMA-ES [Nomura 24]

**Step-size**

**Shape**

$\eta_\Sigma$

single adapted  
learning rate  
for both

**step-size & shape updates:  
implicitly coupled**

**The adapted  $\eta_\Sigma$  cannot independently  
control the step-size & shape updates.**

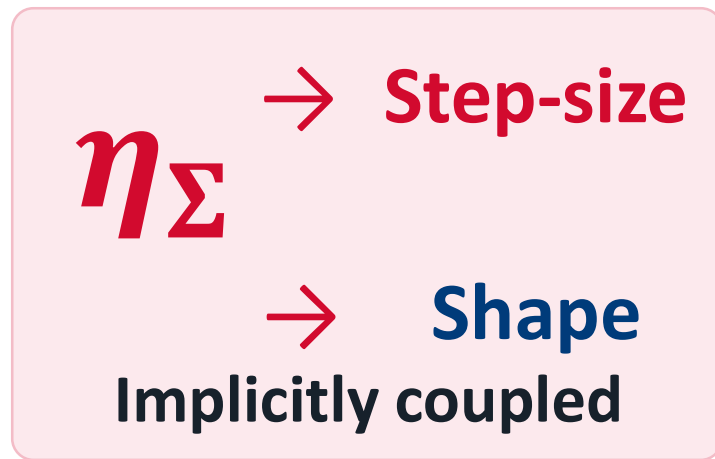
(e.g., cannot accelerate one update & suppress the other at once. )

## Proposed Solution — S3-LRA-xNES:

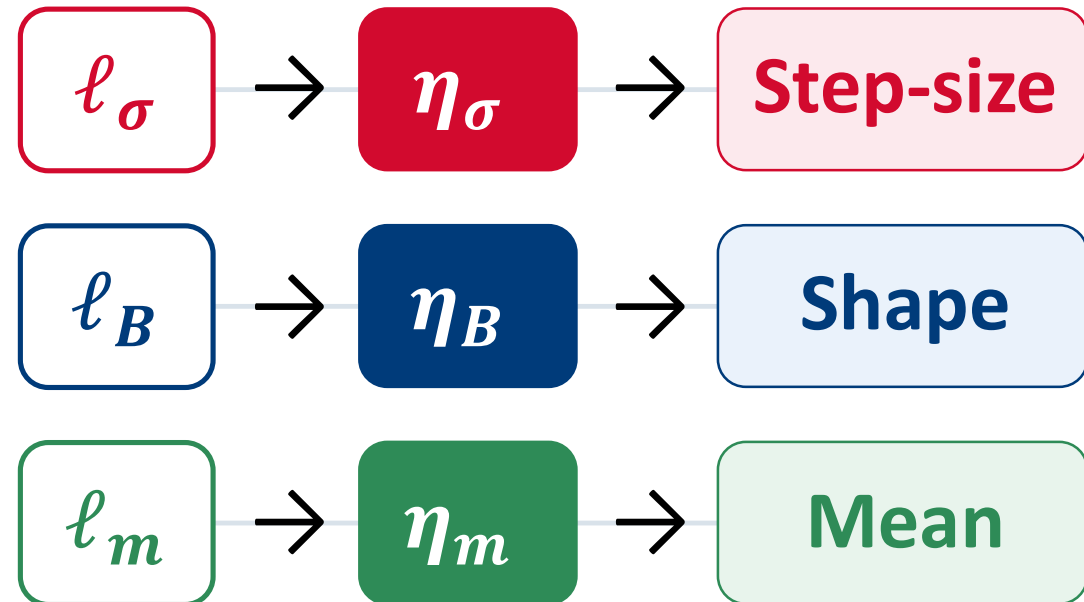
From Shared  $\eta_{\Sigma}$  to Separate Adaptation of Learning Rates

---

LRA-CMA-ES



S3-LRA-xNES



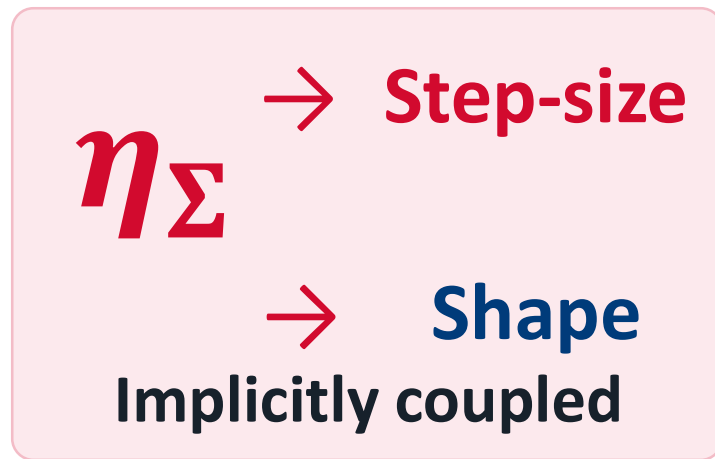
One learning rate can increase while another decreases during the same iteration.

## Proposed Solution — S3-LRA-xNES:

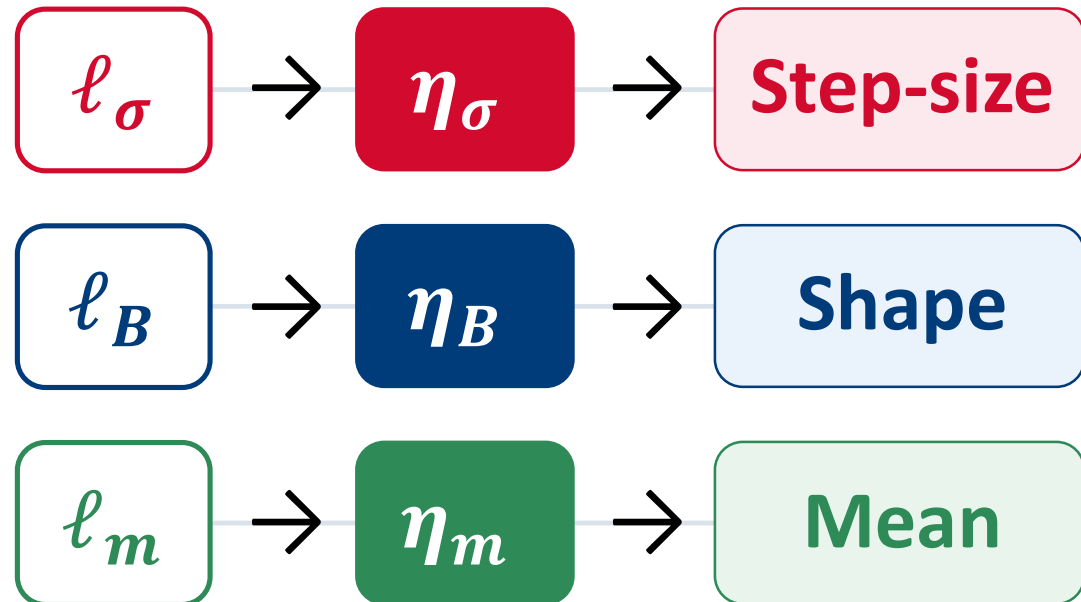
### From Shared $\eta_\Sigma$ to Separate Adaptation of Learning Rates

---

#### LRA-CMA-ES



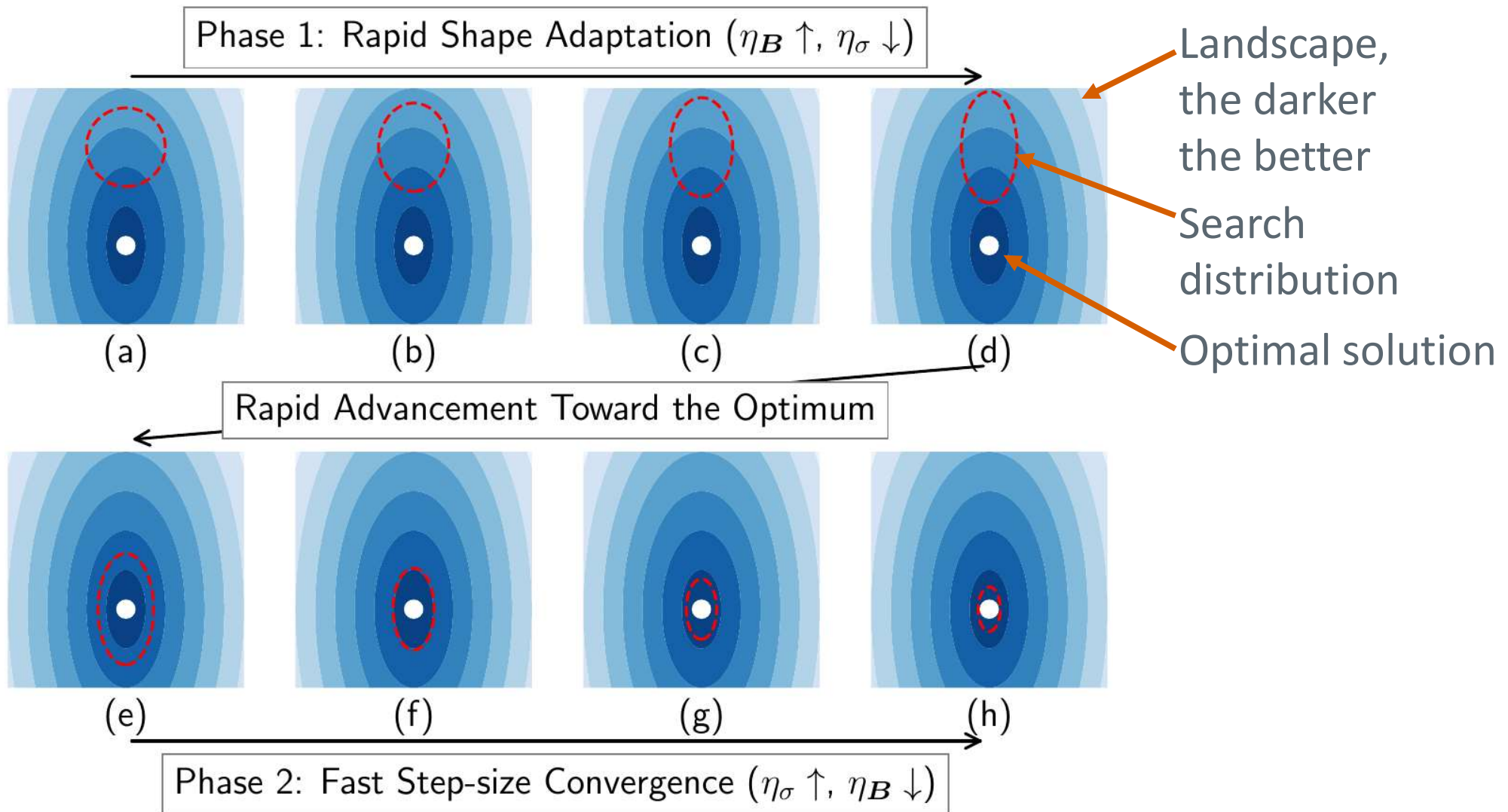
#### S3-LRA-xNES



One learning rate can increase while another decreases during the same iteration.

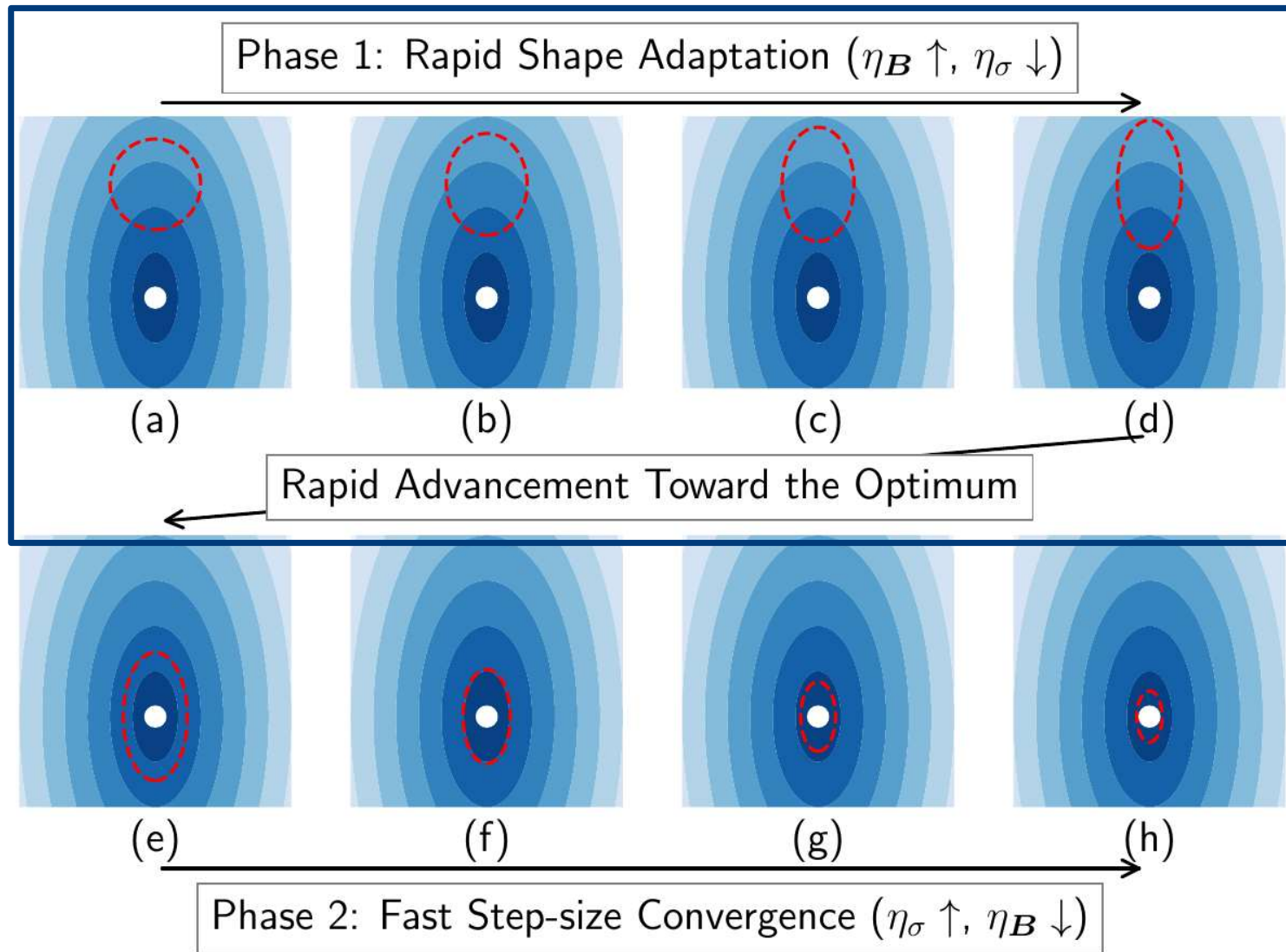
→ This enables “Opposite control of  $\eta_\sigma$  &  $\eta_B$ ”  
described on the next slide.

# Why the Separate Adaptation of Learning rates Is Necessary: Opposite $\eta_\sigma$ & $\eta_B$ Control on Ill-Conditioned Problems





# Why the Separate Adaptation of Learning rates Is Necessary: Opposite $\eta_\sigma$ & $\eta_B$ Control on Ill-Conditioned Problems



## PHASE 1: SHAPE ALIGNMENT

$\eta_B \uparrow$

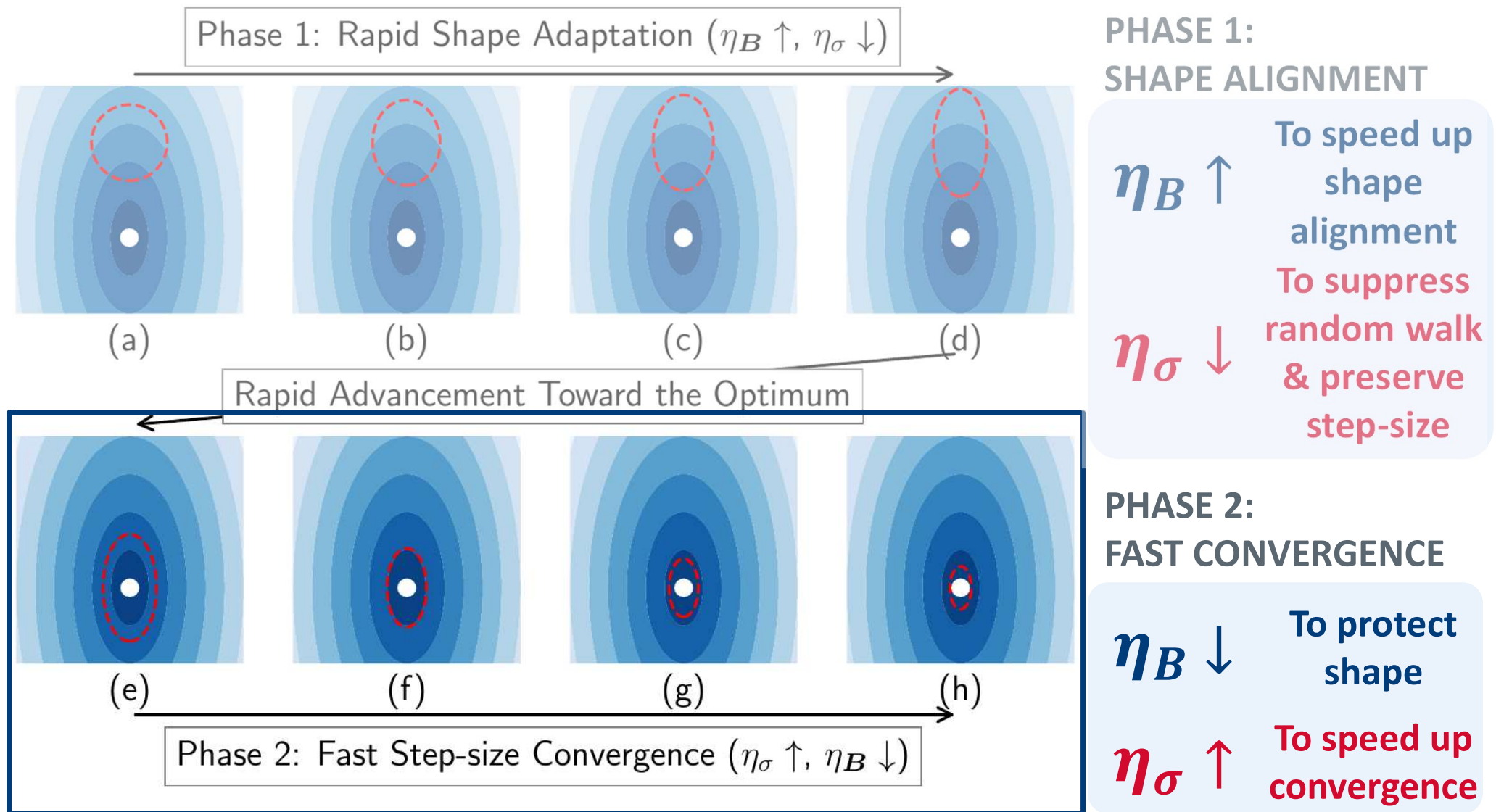
To speed up  
shape  
alignment

$\eta_\sigma \downarrow$

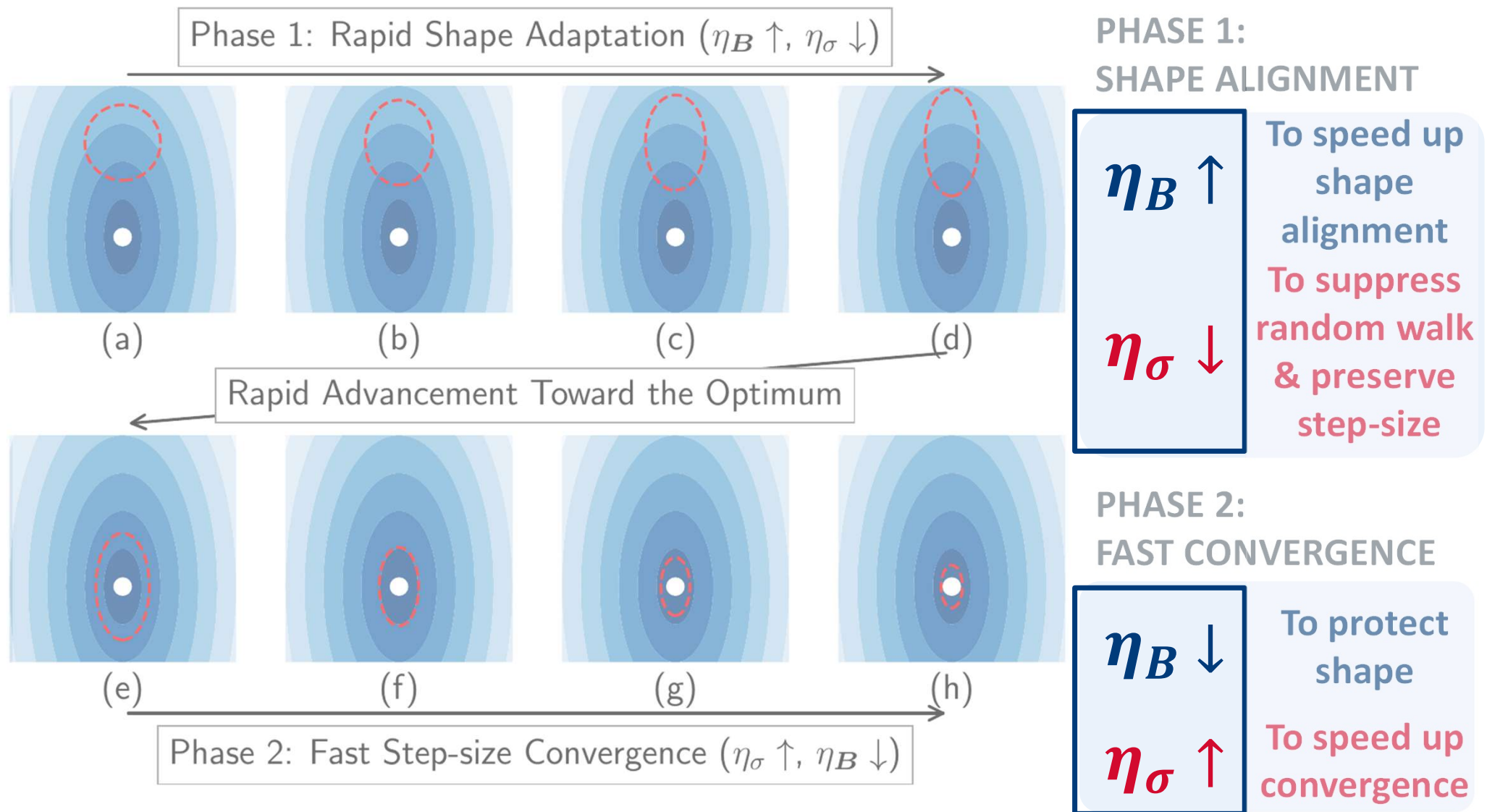
To suppress  
random walk  
& preserve  
step-size



# Why the Separate Adaptation of Learning rates Is Necessary: Opposite $\eta_\sigma$ & $\eta_B$ Control on Ill-Conditioned Problems

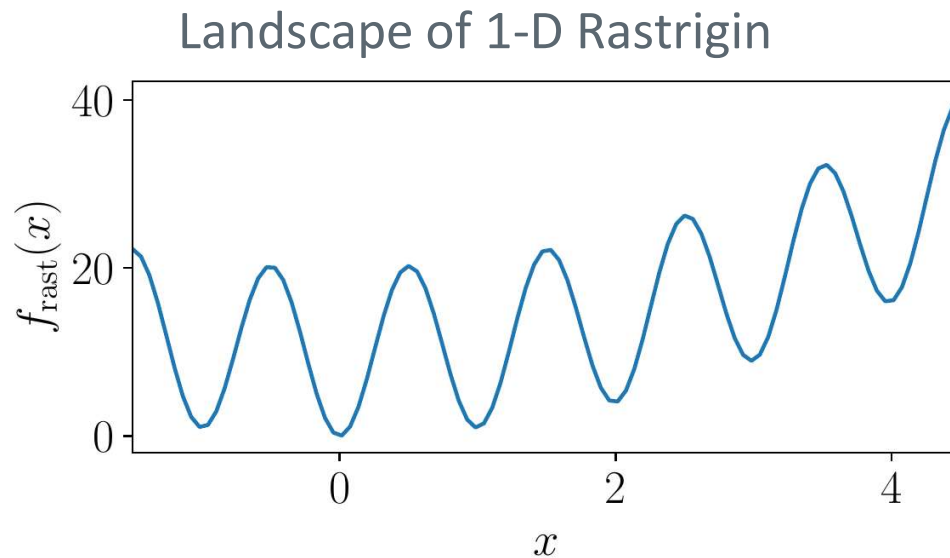
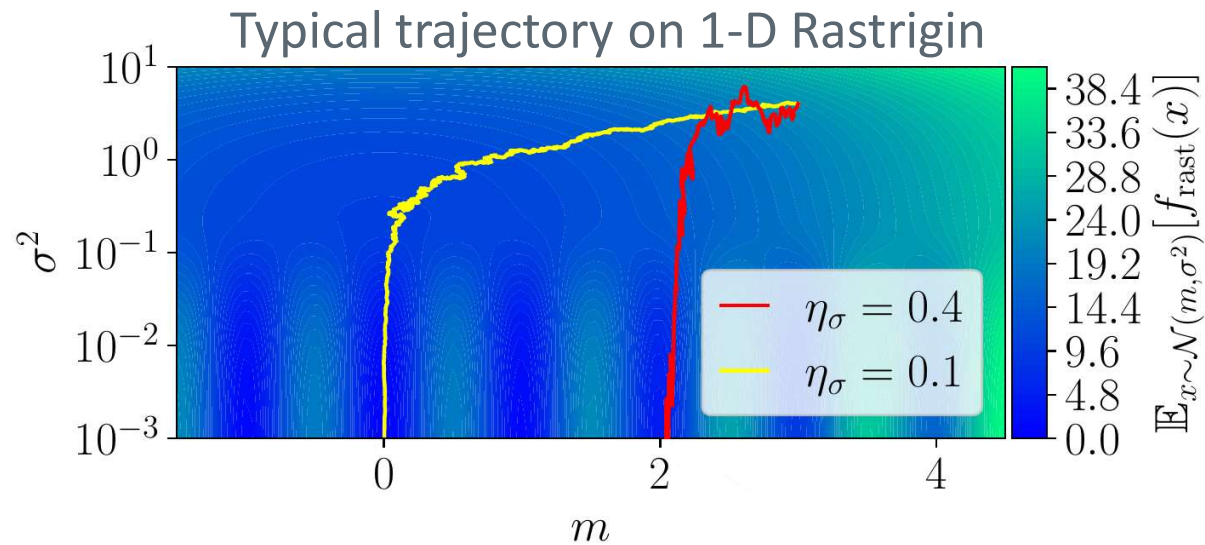


# Why the Separate Adaptation of Learning rates Is Necessary: Opposite $\eta_\sigma$ & $\eta_B$ Control on Ill-Conditioned Problems

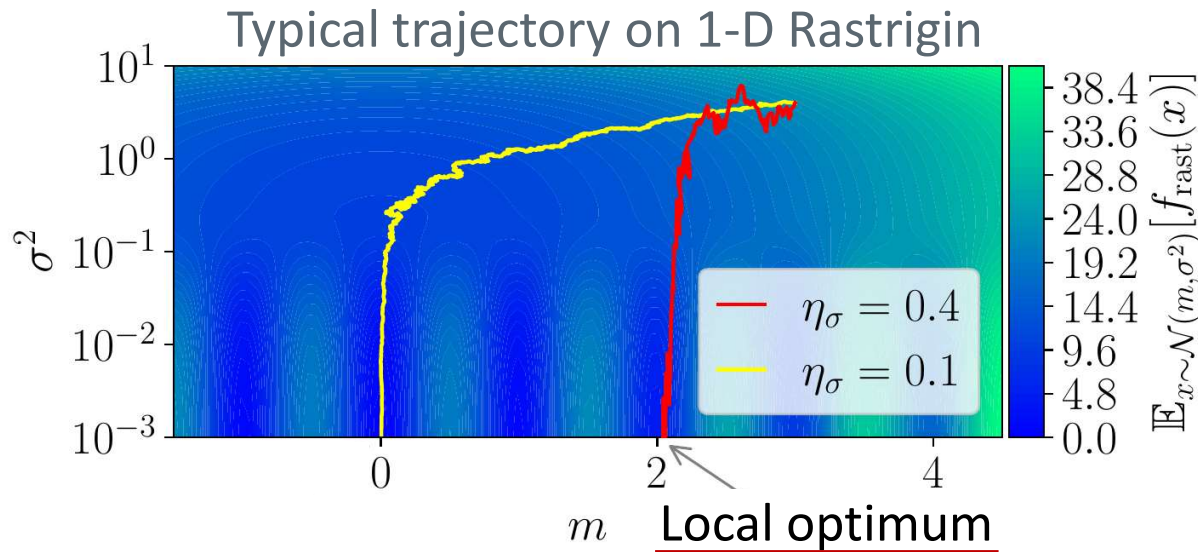


This opposite control is our key novelty:  
It cannot be realized by the shared  $\eta_\Sigma$  in LRA-CMA-ES.

# Why Asymmetric $\eta_\sigma$ Adaptation Is Necessary: Avoiding Local Trapping on Multimodal Problems



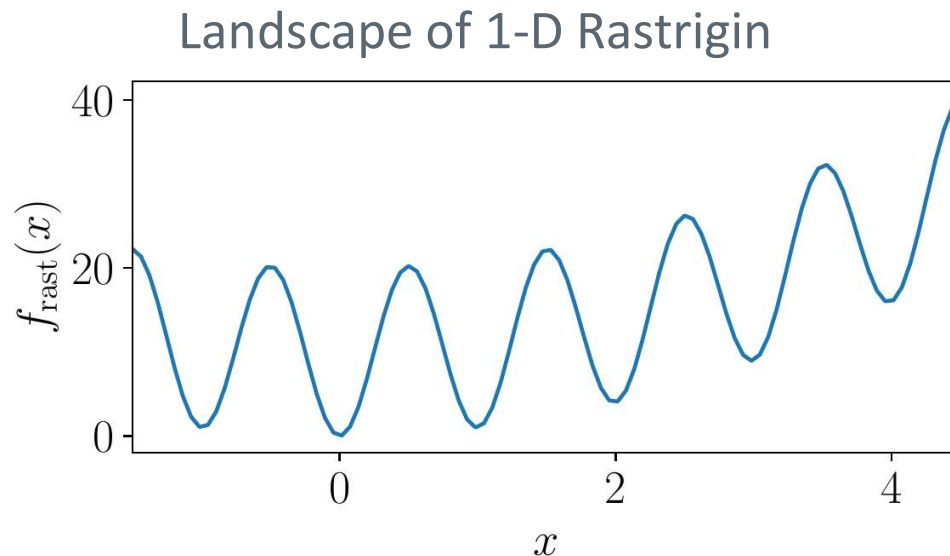
# Why Asymmetric $\eta_\sigma$ Adaptation Is Necessary: Avoiding Local Trapping on Multimodal Problems



With LARGE  $\eta_\sigma$  in red curve

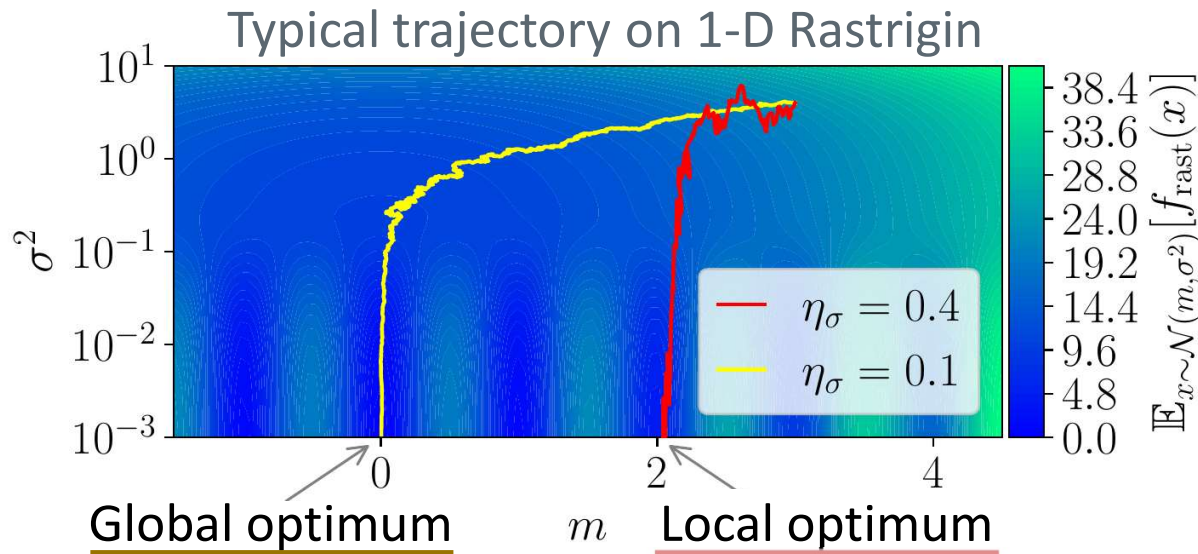
**Aggressive Shrinkage**

trapped at a local optimum





# Why Asymmetric $\eta_\sigma$ Adaptation Is Necessary: Avoiding Local Trapping on Multimodal Problems



With LARGE  $\eta_\sigma$  in red curve

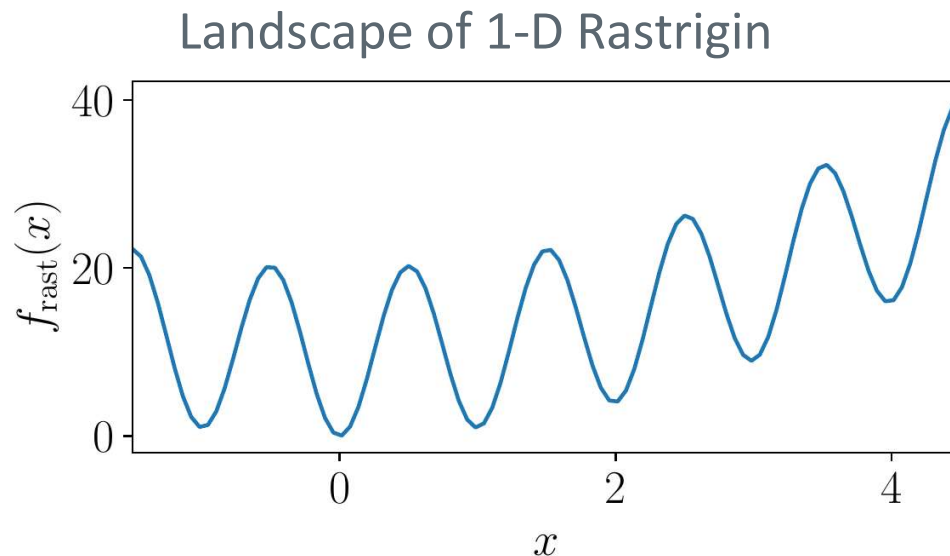
**Aggressive Shrinkage**

trapped at a local optimum

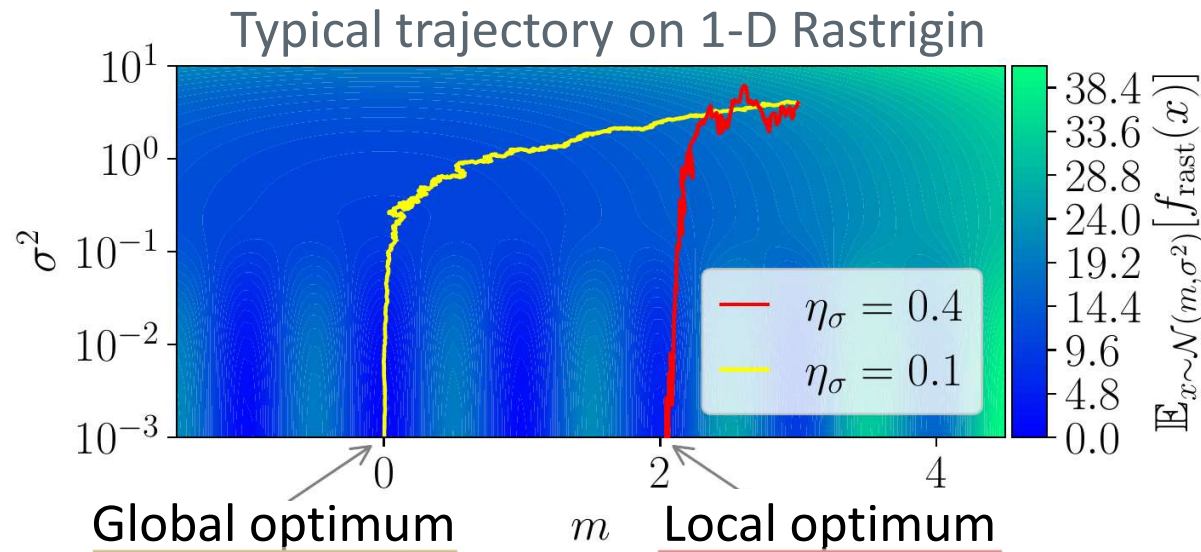
With SMALL  $\eta_\sigma$  in yellow curve

**Cautious Shrinkage**

reaches the global optimum



# Why Asymmetric $\eta_\sigma$ Adaptation Is Necessary: Avoiding Local Trapping on Multimodal Problems



With LARGE  $\eta_\sigma$  in red curve

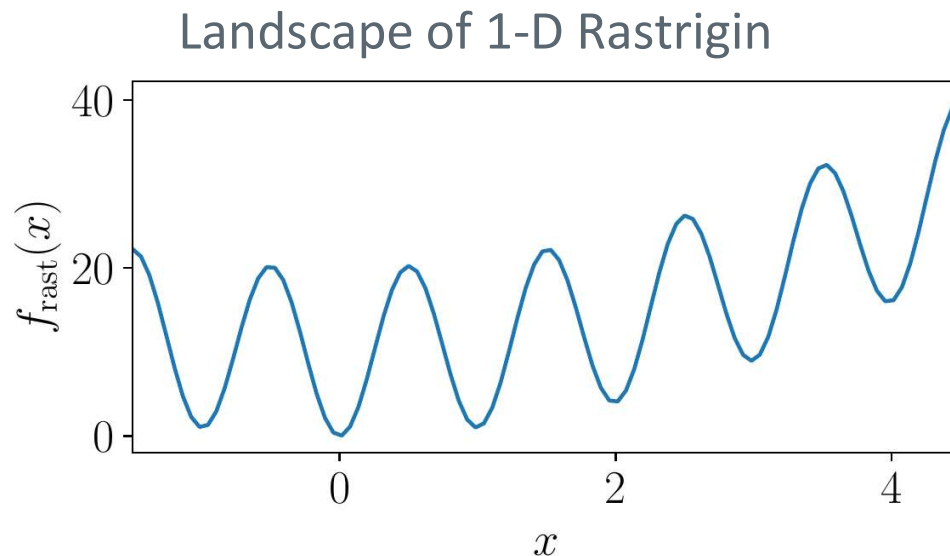
**Aggressive Shrinkage**

trapped at a local optimum

With SMALL  $\eta_\sigma$  in yellow curve

**Cautious Shrinkage**

reaches the global optimum



**Asymmetric  $\eta_\sigma$  Adaptation**

If step-size updates show a random walk:

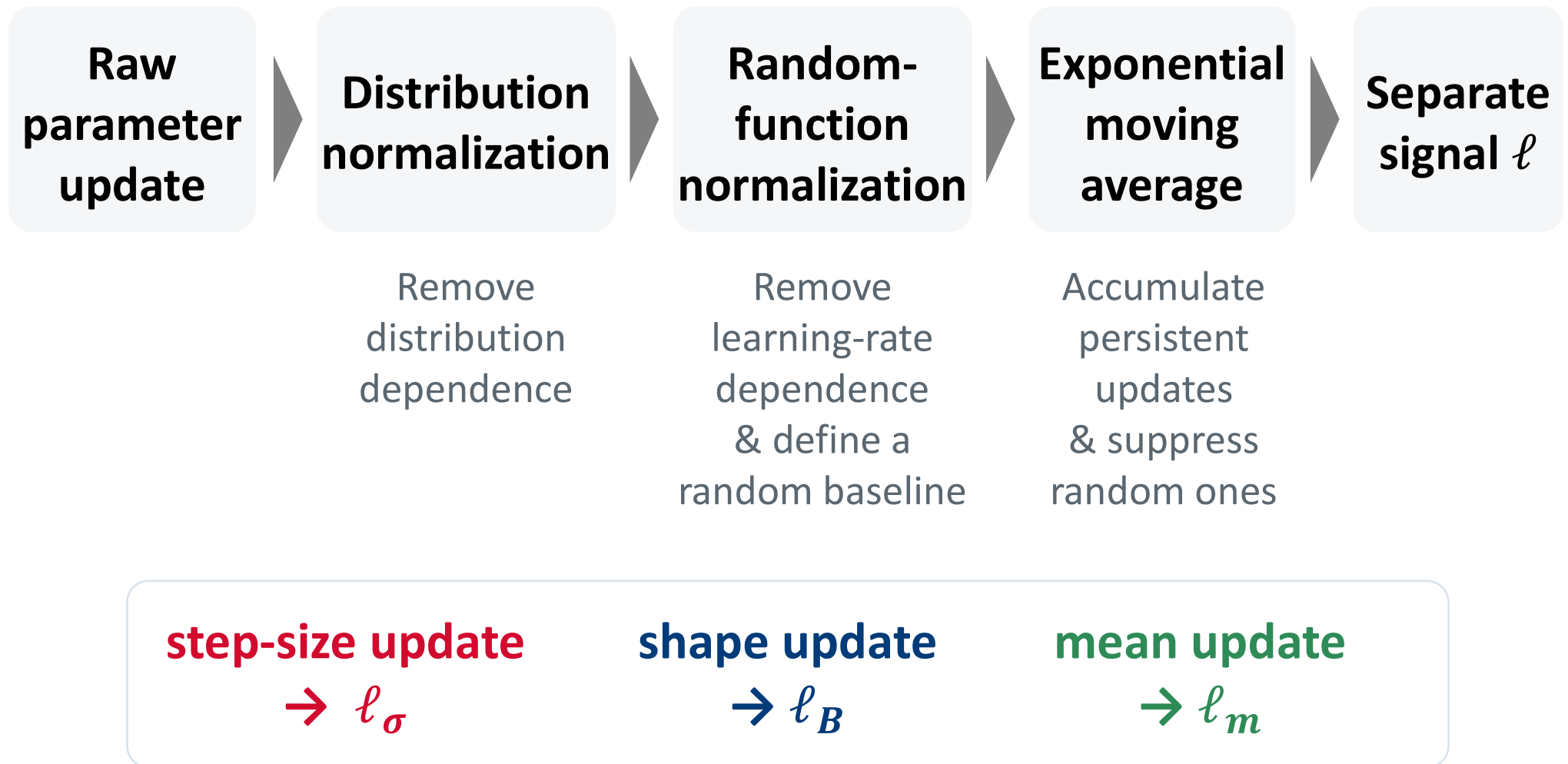
**Decrease  $\eta_\sigma$  quickly**

Otherwise:

**increase it cautiously**

# Signals for Independent Adaptation: Measuring Update Consistencies with Separate Signals

---



$\ell$  is above its random baseline ... directionally consistent updates in corresponding parameter



# Signals for Independent Adaptation: Definition of the Separate Signals

---

$$\tilde{\Delta}_\sigma = \text{Tr} \left( \left( \sqrt{\Sigma(t)}^{-1} \left( \sigma^{(t+1)^2} - \sigma^{(t)^2} \right) B^{(t)} B^{(t)\top} \sqrt{\Sigma(t)}^{-1} \right)^2 \right) / 2,$$

$$\tilde{\Delta}_B = \sqrt{\Sigma(t)}^{-1} \sigma^{(t)^2} \left( B^{(t+1)} B^{(t+1)\top} - B^{(t)} B^{(t)\top} \right) \sqrt{\Sigma(t)}^{-1},$$

$$\tilde{\Delta}_m = \left( \sigma^{(t)} B^{(t)} \right)^{-1} \left( m^{(t+1)} - m^{(t)} \right)$$

**Distribution  
normalization**

$$\Sigma(t) = \sigma^{(t)^2} B^{(t)} B^{(t)\top},$$

$$\begin{aligned} p_B^{(t+1)} &= (1 - c_B) p_B^{(t)} + \sqrt{c_B(2 - c_B)} \tilde{\Delta}_B / \sqrt{\mathbb{E} \left[ \text{Tr} \left( \left( \tilde{\Delta}_B^{\text{rand}} \right)^2 \right) / 2 \right]}, \\ p_m^{(t+1)} &= (1 - c_m) p_m^{(t)} + \sqrt{c_m(2 - c_m)} \tilde{\Delta}_m / \mathbb{E} \left[ \|\tilde{\Delta}_m^{\text{rand}}\| \right], \\ \ell_\sigma^{(t+1)} &= (1 - c_\sigma)^2 \ell_\sigma^{(t)} + c_\sigma(2 - c_\sigma) \tilde{\Delta}_\sigma / \mathbb{E} \left[ \tilde{\Delta}_\sigma^{\text{rand}} \right], \end{aligned}$$

**Exponential  
moving average**

$$\ell_B^{(t+1)} = \text{Tr} \left( p_B^{(t+1)^2} \right) / 2,$$

$$\ell_m^{(t+1)} = \|p_m^{(t+1)}\|,$$

**Random-function  
normalization**

# Detail of the Separated Learning Rate Adaptation: Independent Adaptation and Asymmetric $\eta_\sigma$ Adaptation

for each parameter  $j \in \{\sigma, B, m\}$ :

1. Adaptation evidence

$$d_j = \frac{\ell_j}{\alpha_j} - \gamma_j$$

$\ell_j$ : separate signal,  $\gamma_j$ : random baseline,  
sign of  $d_j$ : (+) increase  $\eta_j$ , (-) decrease it.

2. Independent update

$$\eta_j \leftarrow \text{clip}(\eta_j \exp(\beta_j d_j))$$

$\beta_j$ : update speed,  
clip: allowable range

$\eta_\sigma$

adapted from  $\ell_\sigma$

$\eta_B$

adapted from  $\ell_B$

$\eta_m$

adapted from  $\ell_m$

Asymmetric  $\eta_\sigma$  Adaptation

Two update speeds  $\beta_\sigma^{\text{dec}} > \beta_\sigma^{\text{inc}}$ ;

Decrease  $\eta_\sigma$  quickly, increase it cautiously.

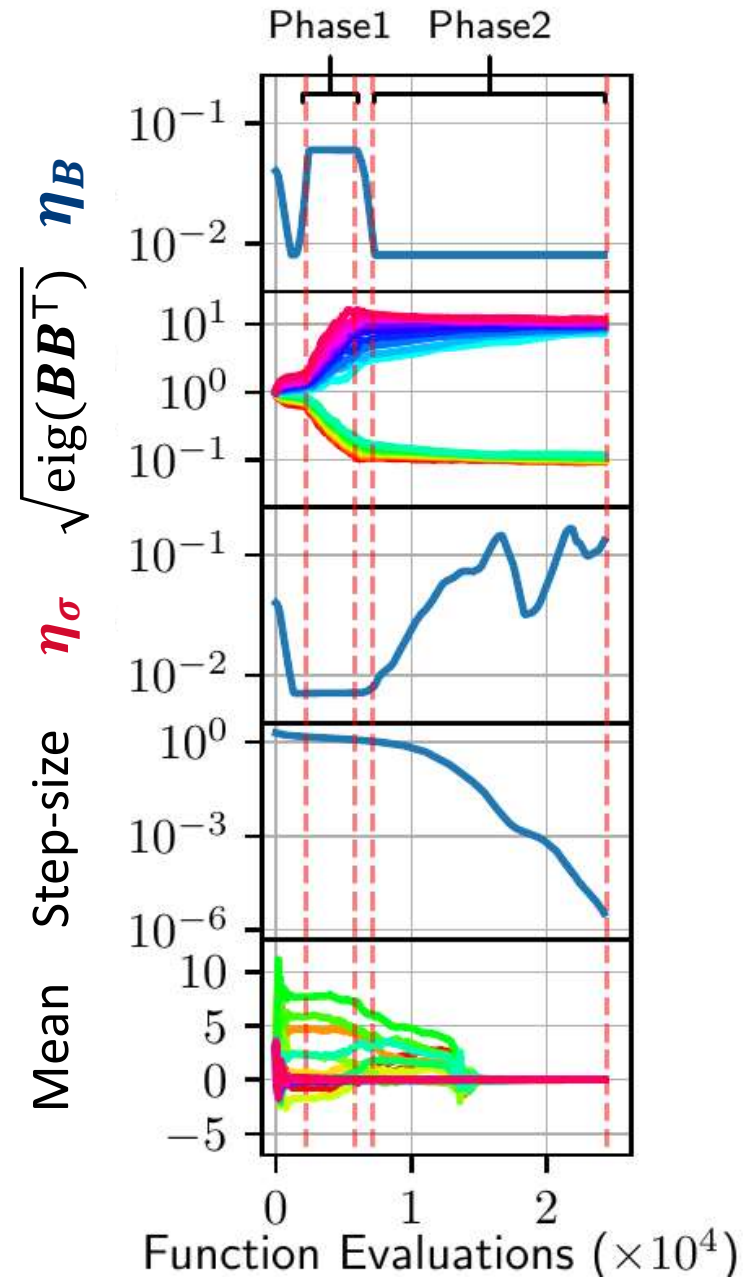
# Research Questions and Evaluation Roadmap: From Dynamics to Performance

---

| Research Question                                                                                                             | Evaluation                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>RQ1: Does the opposite control of <math>\eta_\sigma</math> &amp; <math>\eta_B</math> occur in the two intended phases?</b> | Learning rate dynamics on <u><math>k</math>-Tablet</u><br>Ill-conditioned unimodal |
| <b>RQ2: Does the asymmetric <math>\eta_\sigma</math> adaptation prevent trapping in a local optimum?</b>                      | Ablation on <u>Bohachevsky</u><br>Multimodal                                       |
| <b>RQ3: Does S3-LRA-xNES improve speed and robustness?</b>                                                                    | Benchmark comparison across five functions, 20, 40, & 80 dimensions, & 50 trials.  |

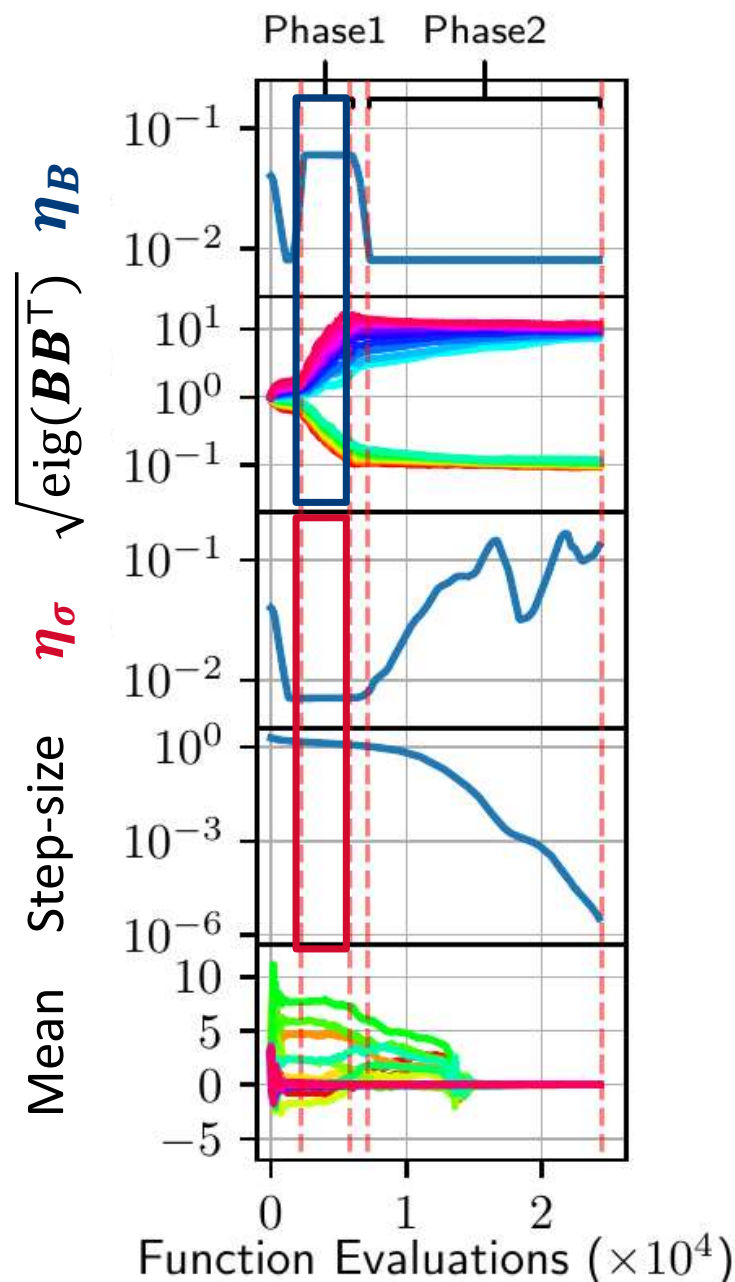
# Answer to RQ1:Independent Adaptation Realizes the Intended Two-Phase Controls

Representative run on 20-D k-Tablet



# Answer to RQ1:Independent Adaptation Realizes the Intended Two-Phase Controls

Representative run on 20-D k-Tablet



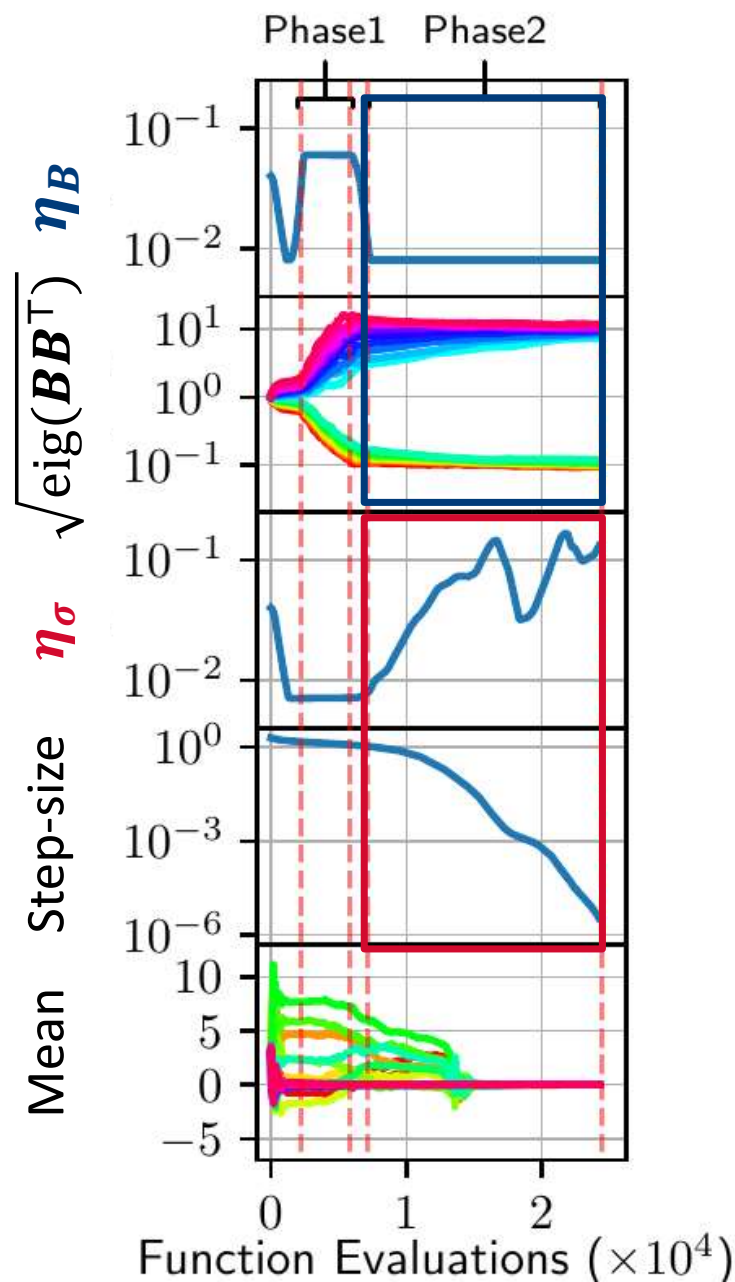
## PHASE 1

$\eta_B \uparrow$  shape aligned quickly

$\eta_\sigma \downarrow$  step-size was preserved

# Answer to RQ1:Independent Adaptation Realizes the Intended Two-Phase Controls

Representative run on 20-D k-Tablet



## PHASE 1

$\eta_B \uparrow$  shape aligned quickly

$\eta_\sigma \downarrow$  step-size was preserved

## PHASE 2

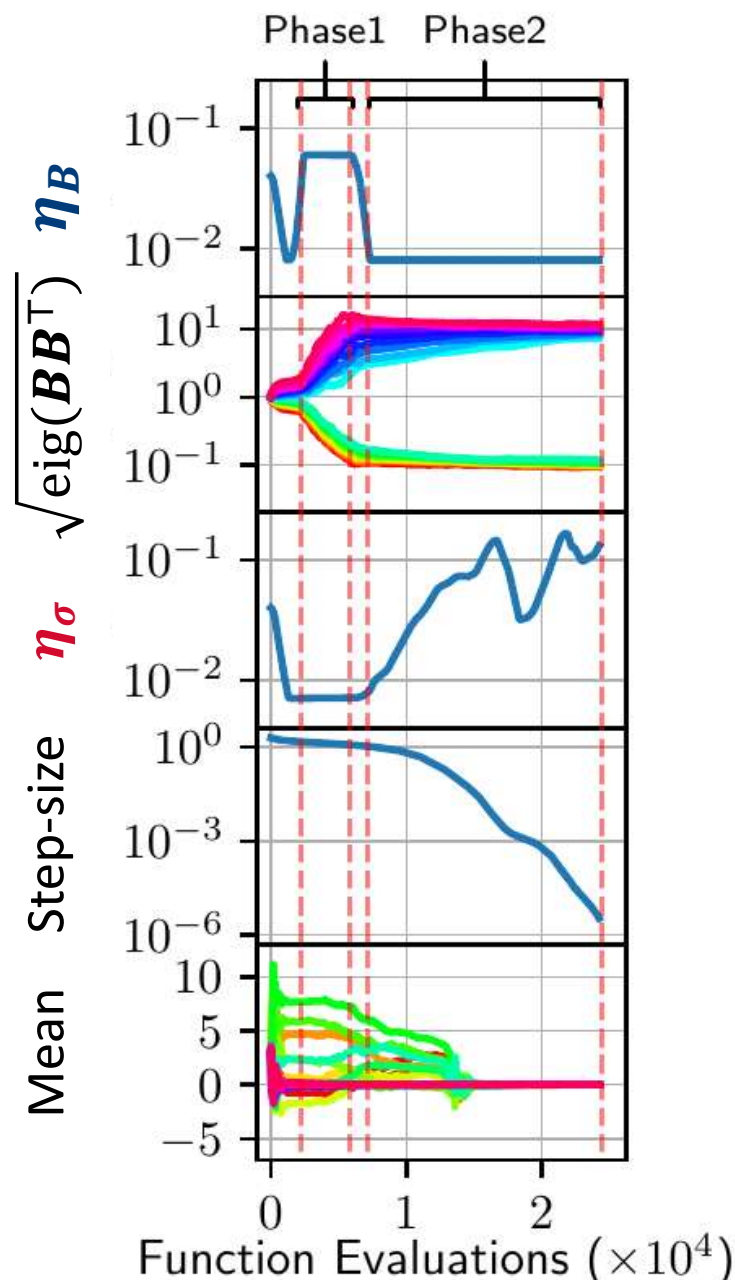
$\eta_B \downarrow$  shape was preserved

$\eta_\sigma \uparrow$  step-size shrunk quickly



# Answer to RQ1:Independent Adaptation Realizes the Intended Two-Phase Controls

Representative run on 20-D k-Tablet



PHASE 1

$\eta_B \uparrow$  shape aligned quickly  
 $\eta_\sigma \downarrow$  step-size was preserved

PHASE 2

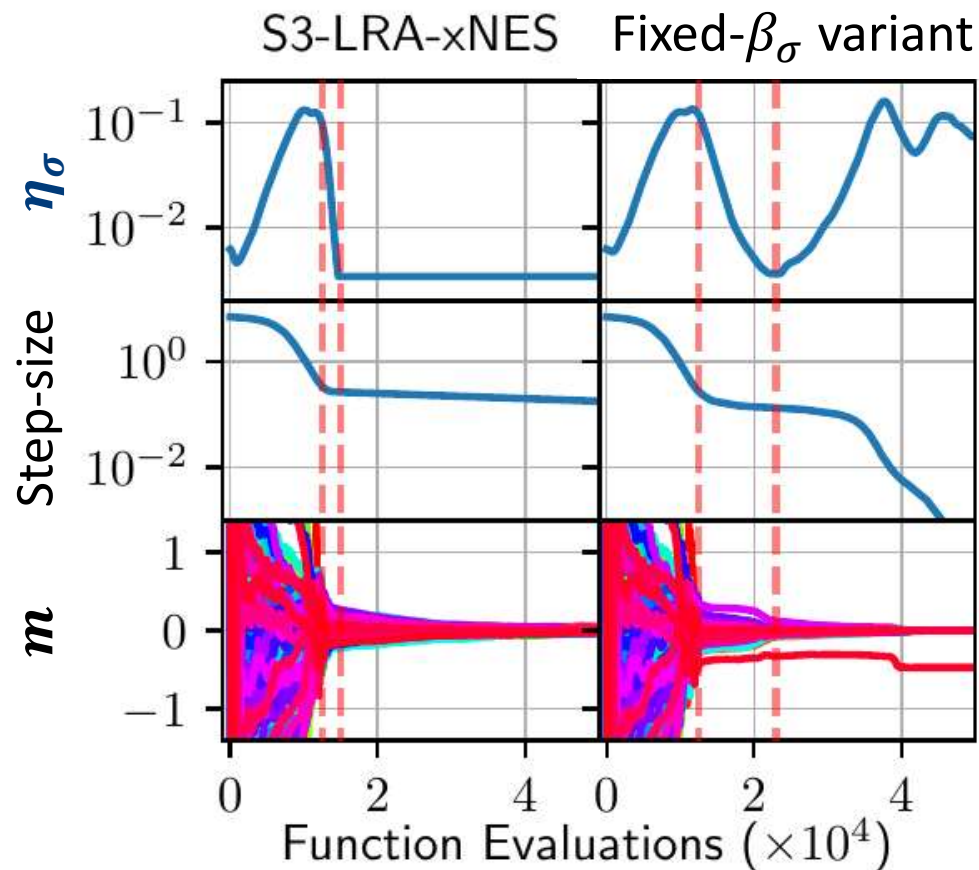
$\eta_B \downarrow$  shape was preserved  
 $\eta_\sigma \uparrow$  step-size shrunk quickly

The separated learning rate adaptation recovers the required opposite control.

# Answer to RQ2: Ablation Shows That Asymmetric $\eta_\sigma$ Adaptation Prevents Local Trapping

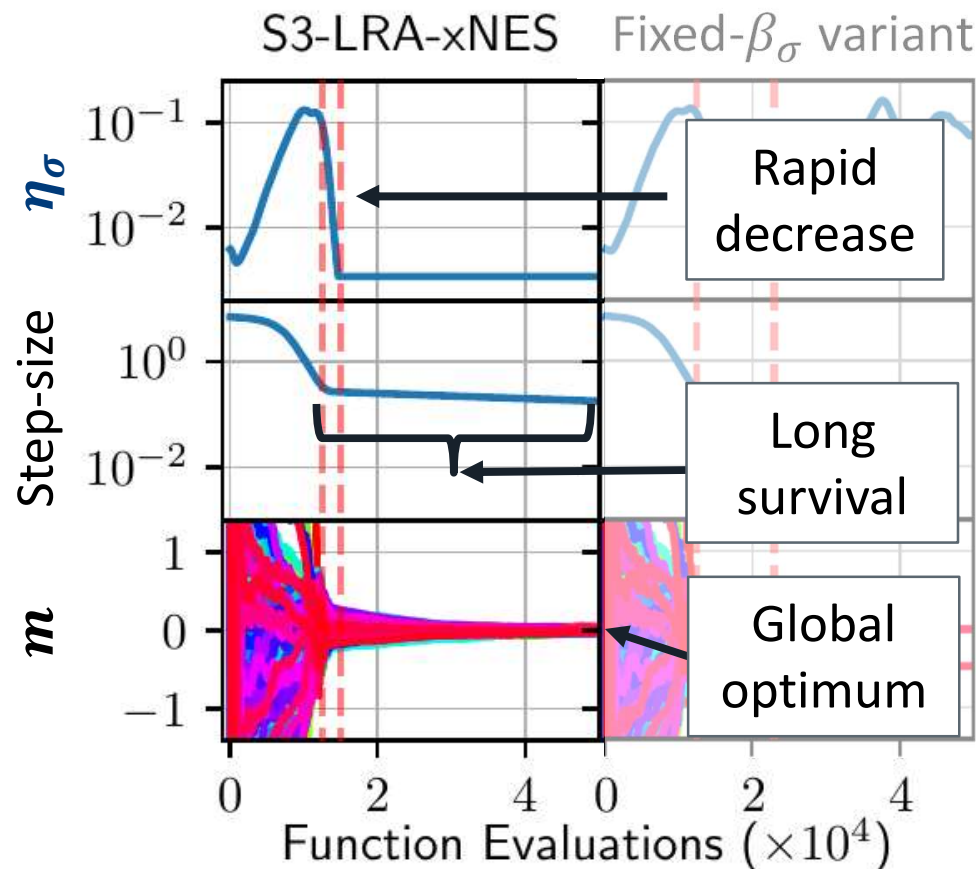
---

Ablation of the asymmetric  $\eta_\sigma$  adaptation on 80-D Bohachevsky



# Answer to RQ2: Ablation Shows That Asymmetric $\eta_\sigma$ Adaptation Prevents Local Trapping

Ablation of the asymmetric  $\eta_\sigma$  adaptation on 80-D Bohachevsky



## S3-LRA-xNES

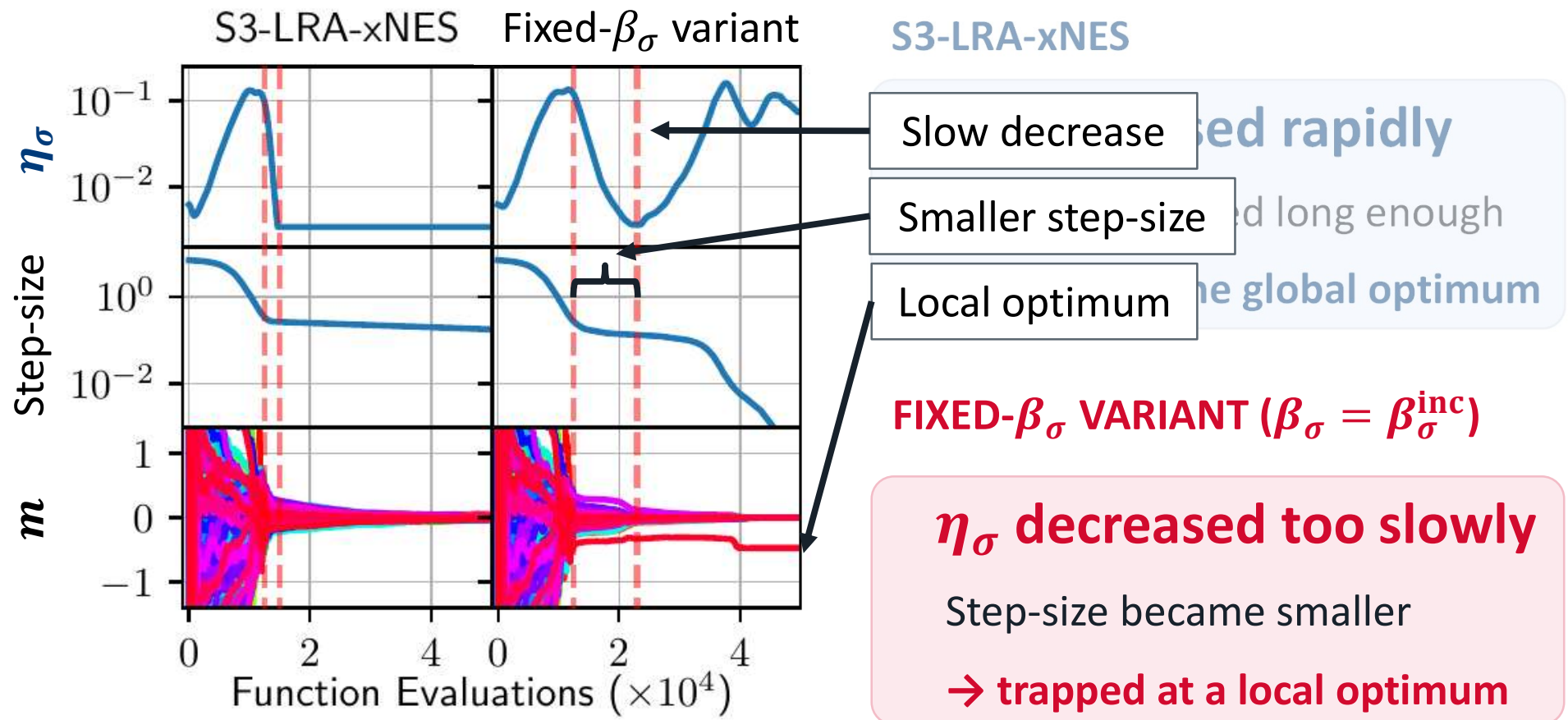
**$\eta_\sigma$  decreased rapidly**

Step-size survived long enough

**→ reached the global optimum**

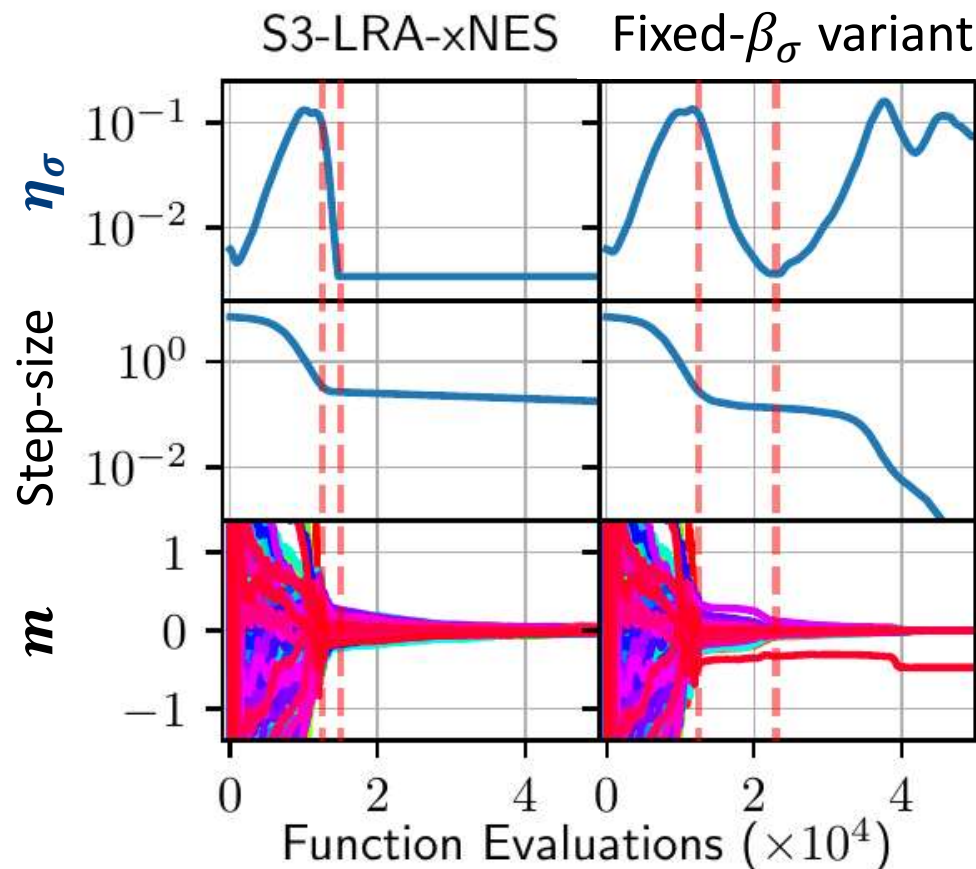
# Answer to RQ2: Ablation Shows That Asymmetric $\eta_\sigma$ Adaptation Prevents Local Trapping

Ablation of the asymmetric  $\eta_\sigma$  adaptation on 80-D Bohachevsky



# Answer to RQ2: Ablation Shows That Asymmetric $\eta_\sigma$ Adaptation Prevents Local Trapping

Ablation of the asymmetric  $\eta_\sigma$  adaptation on 80-D Bohachevsky



## S3-LRA-xNES

**$\eta_\sigma$  decreased rapidly**

Step-size survived long enough

→ reached the global optimum

## FIXED- $\beta_\sigma$ VARIANT ( $\beta_\sigma = \beta_\sigma^{\text{inc}}$ )

**$\eta_\sigma$  decreased too slowly**

Step-size became smaller

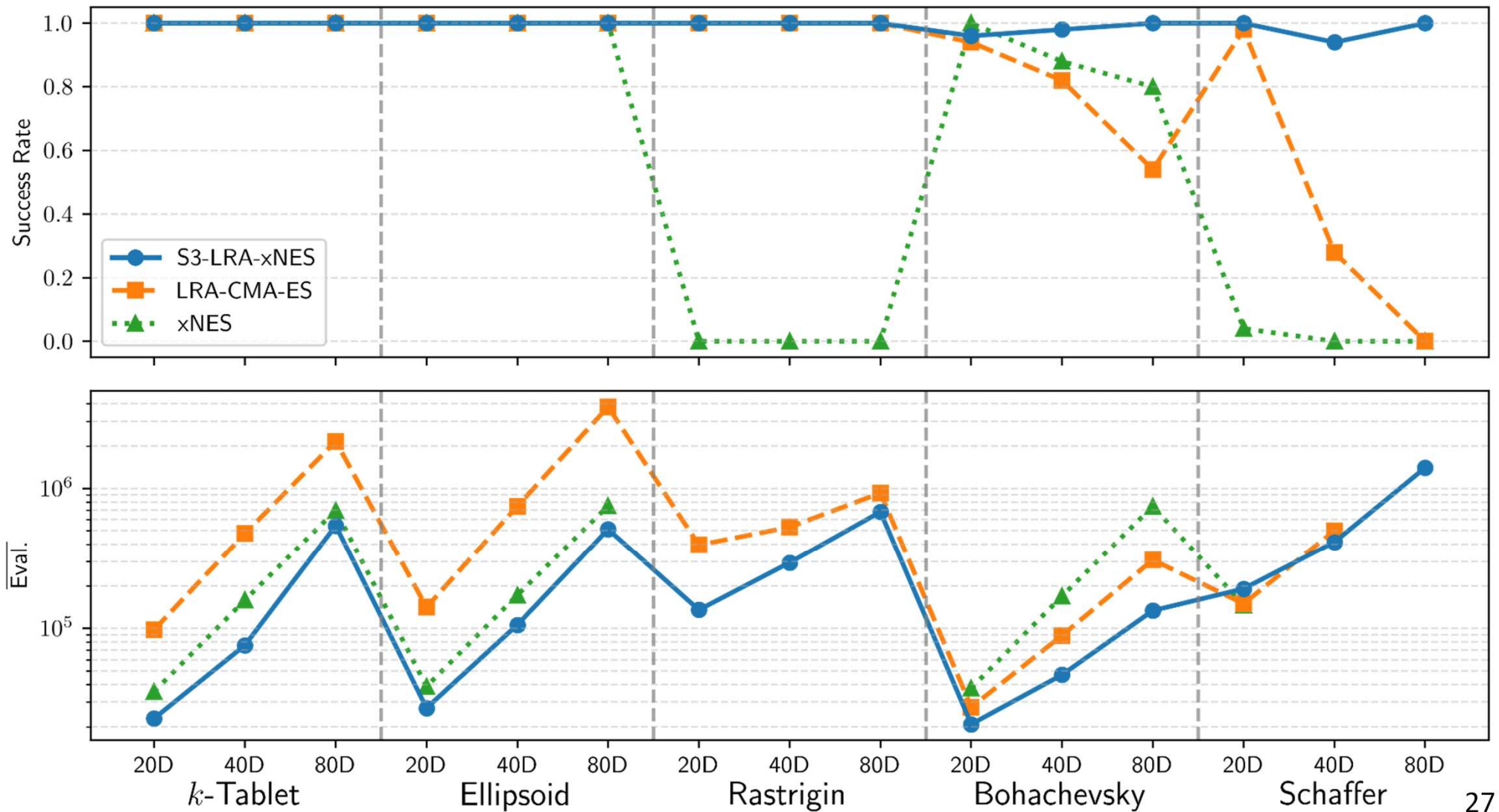
→ trapped at a local optimum

This ablation suggests the effectiveness of the asymmetric  $\eta_\sigma$  adaptation.



# Answer to RQ3: Faster on Ill-Conditioned Problems, More Robust on Multimodal Problems

Success Rates across 50 trials & average evaluations over successful trials ( $\overline{\text{Eval.}}$ )

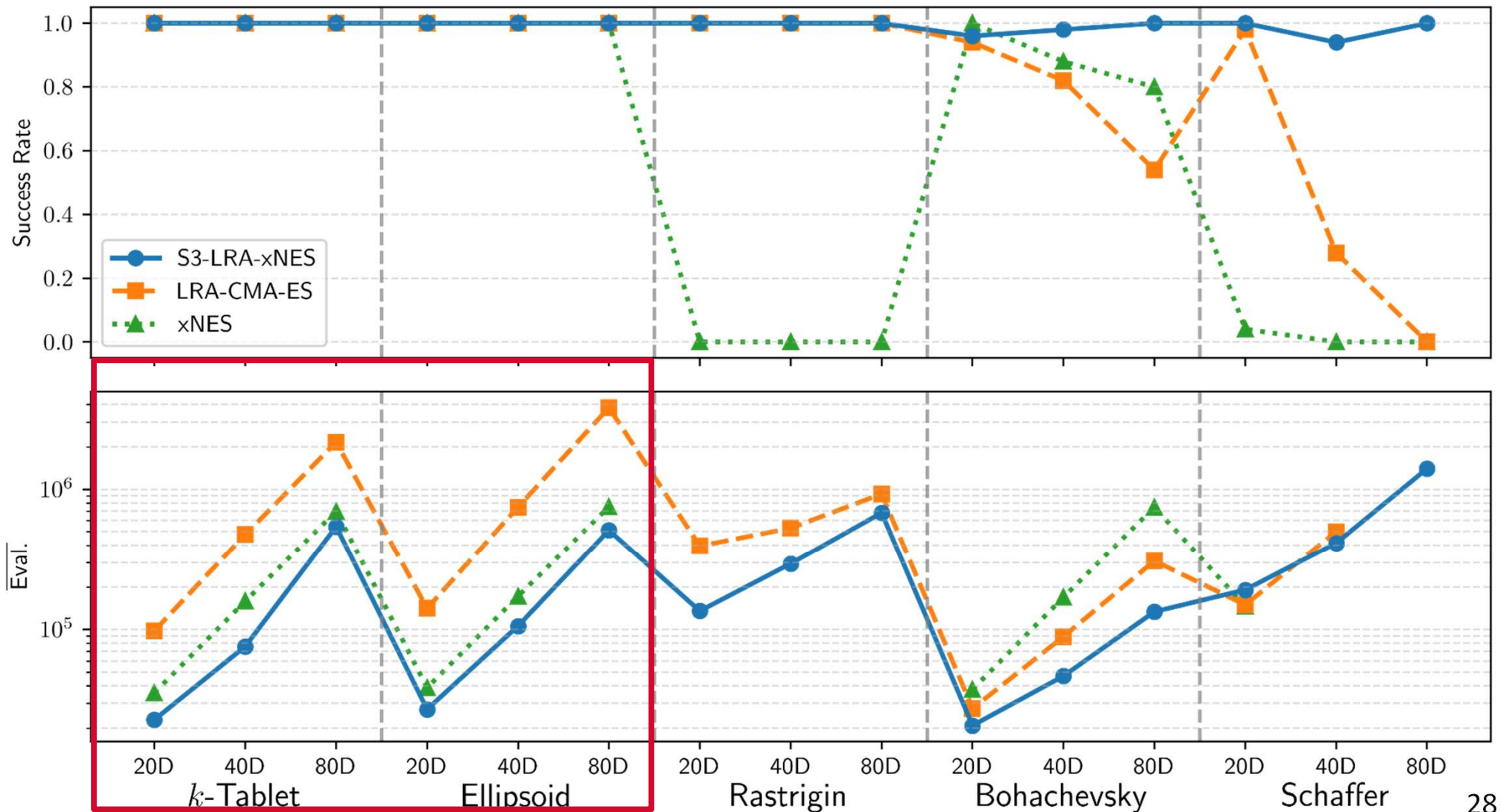




# Answer to RQ3: Faster on Ill-Conditioned Problems, More Robust on Multimodal Problems

Success Rates across 50 trials & average evaluations over successful trials ( $\overline{\text{Eval.}}$ )

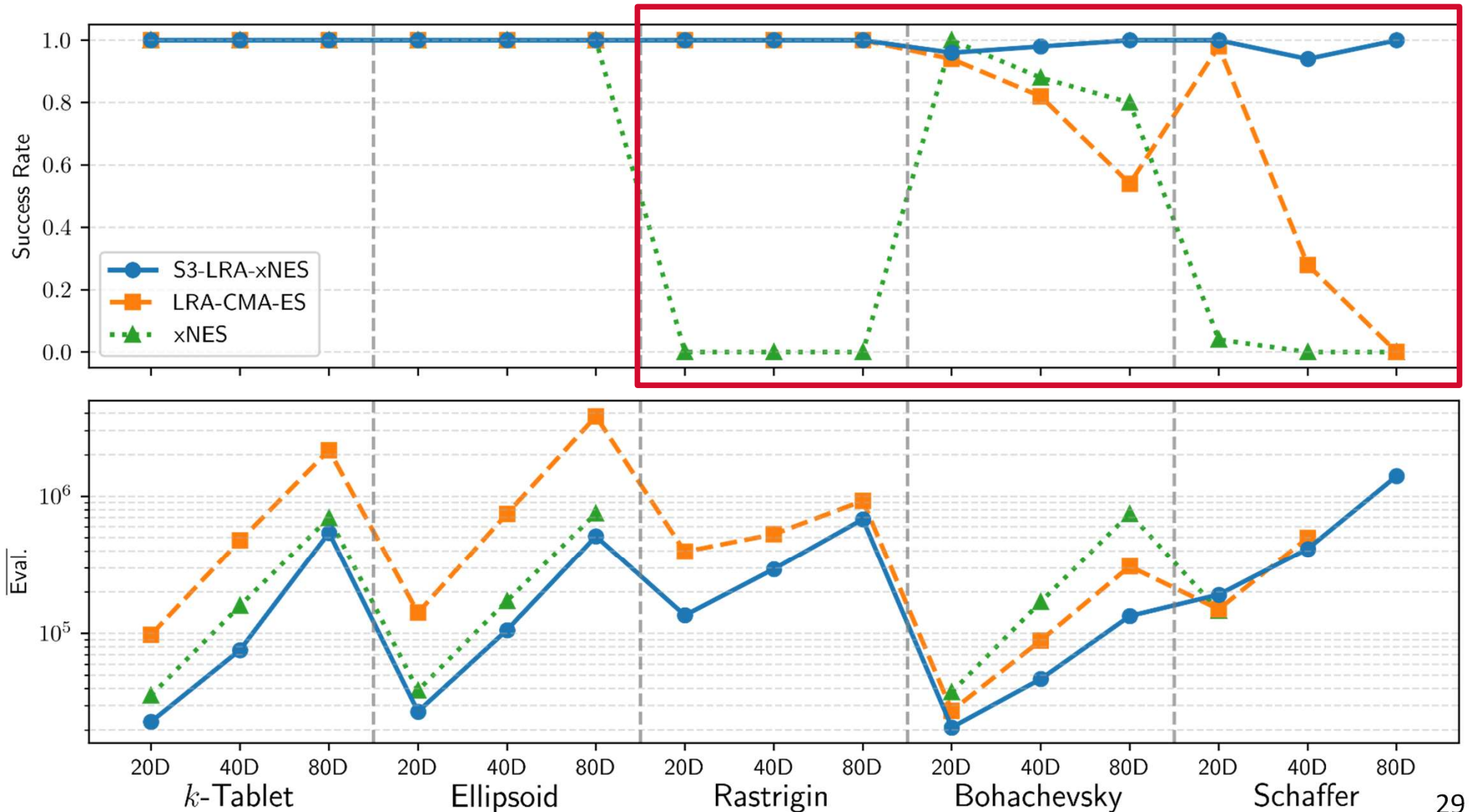
**Ill-Conditioned: fastest in all 6 settings;**  
**-82.4% to LRA-CMA-ES & -33.8% to xNES**



# Answer to RQ3: Faster on Ill-Conditioned Problems, More Robust on Multimodal Problems

Success Rates across 50 trials & average evaluations over successful trials ( $\overline{\text{Eval.}}$ )

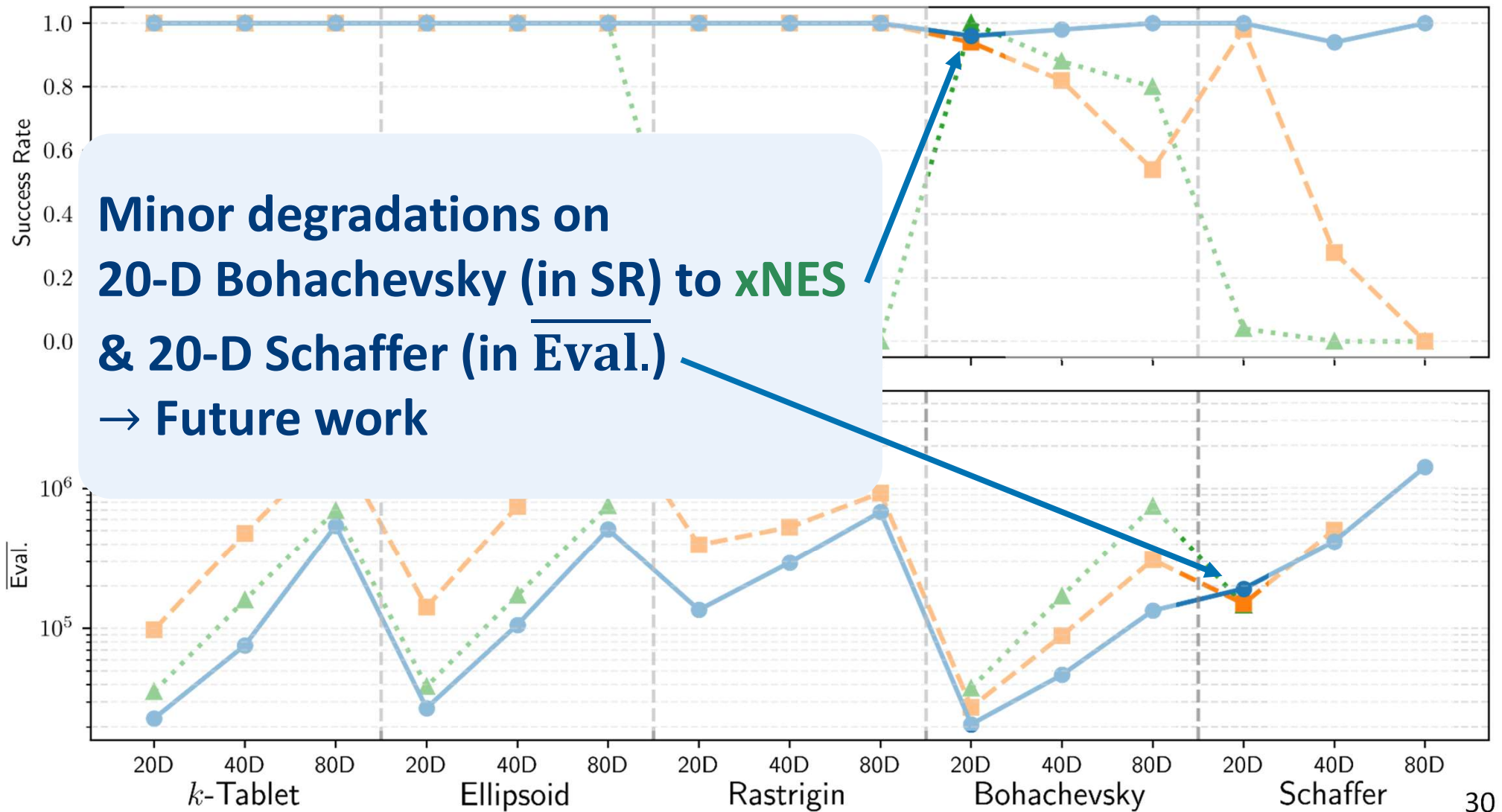
**Ill-Conditioned: fastest in all 6 settings; -82.4% to LRA-CMA-ES & -33.8% to xNES**      **Multimodal: improved SRs; 50/50 on 80-D problems**



# Answer to RQ3: Faster on Ill-Conditioned Problems, More Robust on Multimodal Problems

Success Rates across 50 trials & average evaluations over successful trials ( $\overline{\text{Eval.}}$ )

Ill-Conditioned: fastest in all 6 settings;    Multimodal: improved SRs;  
-82.4% to LRA-CMA-ES & -33.8% to xNES    50/50 on 80-D problems



# Conclusion

## SUMMARY

- 1 S3-LRA-xNES adapts  $\eta_\sigma$ ,  $\eta_B$ , &  $\eta_m$  independently
- 2 Opposite control of  $\eta_\sigma$  &  $\eta_B$  accelerated optimization on ill-conditioned problems
- 3 Asymmetric  $\eta_\sigma$  adaptation improved robustness on multimodal problems
- 4 Main Results **-82.4%** & **50/50** suggest effectiveness of S3-LRA-xNES

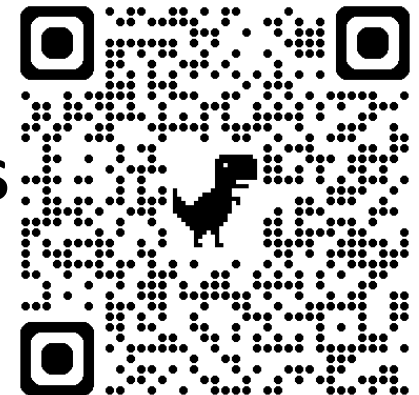
## FUTURE WORK

More principled and automated hyperparameter adaptation

Theoretical understanding of the adaptation mechanisms

Investigate the two minor 20-D degradations

S3-LRA-xNES  
code:



# References

---

[Nomura 24] Masahiro Nomura, Youhei Akimoto, and Isao Ono. CMA-ES with Learning Rate Adaptation. *ACM Trans. Evol. Learn. Optim.*, September 2024.

[Glasmachers 10] Tobias Glasmachers, Tom Schaul, Sun Yi, Daan Wierstra, and Jürgen Schmidhuber. Exponential natural evolution strategies. In *Proceedings of the 12th annual conference on Genetic and evolutionary computation*, pages 393–400, 2010.

# Appendix

---



# Details of the Benchmark Comparison in RQ3

Definitions of the benchmark problems and their initial distributions used in the comparison.  $d$  denotes problem dimension.

| Definitions                                                                                                                                                       | Initial Distributions     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| $f_{k\text{-Tablet}}(\mathbf{x}) = \sum_{i=1}^k x_i^2 + \sum_{i=k+1}^d (100x_i)^2$                                                                                | $\mathcal{N}(3, 2^2 I)$   |
| $f_{\text{Ellipsoid}}(\mathbf{x}) = \sum_{i=1}^d \left( 1000^{\frac{i-1}{d-1}} x_i \right)^2$                                                                     | $\mathcal{N}(3, 2^2 I)$   |
| $f_{\text{Rastrigin}}(\mathbf{x}) = 10d + \sum_{i=1}^d (x_i^2 - 10 \cos(2\pi x_i))$                                                                               | $\mathcal{N}(3, 2^2 I)$   |
| $f_{\text{Bohachevsky}}(\mathbf{x}) = \sum_{i=1}^{d-1} (x_i^2 + 2x_{i+1}^2 - 0.3 \cos(3\pi x_i) - 0.4 \cos(4\pi x_{i+1}) + 0.7)$                                  | $\mathcal{N}(8, 7^2 I)$   |
| $f_{\text{Schaffer}}(\mathbf{x}) = \sum_{i=1}^{d-1} (x_i^2 + x_{i+1}^2)^{0.25} \cdot \left[ \sin^2 \left( 50 \cdot (x_i^2 + x_{i+1}^2)^{0.1} \right) + 1 \right]$ | $\mathcal{N}(55, 45^2 I)$ |

\* All optimal solutions are at 0.

# Details of the Benchmark Comparison in RQ3 (cont.)

---

Setups:

- $k = d/2$  for the  $k$ -Tablet function.
- All algorithms used their default hyperparameter settings and the same population size  $\lambda = 4 + \lfloor 3 \log(d) \rfloor$ ; see the next slide and the original publications [Nomura 24, Glasmachers 10] for details.
- A trial is considered as successful if a solution achieves a target function value of  $10^{-8}$ .
- A trial is considered a failure if either the evaluation budget,  $10^7$  for  $d = 20, 40$ , and  $2 \cdot 10^7$  for  $d = 80$ , is exceeded, or if the multivariate normal distribution degenerates, i.e., the covariance matrix contains negative eigenvalues.

# Details of the Benchmark Comparison in RQ3 (cont.)

| benchmark problem | dimension | S3-LRA-xNES |                                         | LRA-CMA-ES |                                         | xNES  |                                         |
|-------------------|-----------|-------------|-----------------------------------------|------------|-----------------------------------------|-------|-----------------------------------------|
|                   |           | SR          | $\overline{\text{Eval.}} (\times 10^3)$ | SR         | $\overline{\text{Eval.}} (\times 10^3)$ | SR    | $\overline{\text{Eval.}} (\times 10^3)$ |
| <i>k</i> -Tablet  | 20        | 50/50       | 22.94 (1.36)                            | 50/50      | 98.01 (5.93)                            | 50/50 | 35.65 (0.36)                            |
|                   | 40        | 50/50       | 76.05 (3.47)                            | 50/50      | 476.48 (19.01)                          | 50/50 | 159.16 (0.90)                           |
|                   | 80        | 50/50       | 543.18 (5.20)                           | 50/50      | 2165.02 (64.00)                         | 50/50 | 697.19 (2.35)                           |
| Ellipsoid         | 20        | 50/50       | 27.19 (1.99)                            | 50/50      | 141.40 (9.15)                           | 50/50 | 38.69 (0.42)                            |
|                   | 40        | 50/50       | 106.08 (5.37)                           | 50/50      | 747.04 (30.05)                          | 50/50 | 172.40 (0.82)                           |
|                   | 80        | 50/50       | 510.66 (9.30)                           | 50/50      | 3795.83 (76.12)                         | 50/50 | 752.12 (1.93)                           |
| Rastrigin         | 20        | 50/50       | 135.48 (6.69)                           | 50/50      | 394.84 (70.03)                          | 0/50  | 0.00 (0.00)                             |
|                   | 40        | 50/50       | 294.50 (6.89)                           | 50/50      | 529.79 (111.93)                         | 0/50  | 0.00 (0.00)                             |
|                   | 80        | 50/50       | 680.69 (9.93)                           | 50/50      | 927.19 (162.34)                         | 0/50  | 0.00 (0.00)                             |
| Bohachevsky       | 20        | 48/50       | 20.87 (2.25)                            | 47/50      | 27.54 (1.61)                            | 50/50 | 37.65 (0.35)                            |
|                   | 40        | 49/50       | 46.82 (6.24)                            | 41/50      | 89.17 (4.59)                            | 44/50 | 169.77 (0.73)                           |
|                   | 80        | 50/50       | 133.91 (10.75)                          | 27/50      | 307.81 (19.85)                          | 40/50 | 747.30 (2.12)                           |
| Schaffer          | 20        | 50/50       | 191.58 (9.96)                           | 49/50      | 149.90 (4.41)                           | 2/50  | <b>146.31 (0.79)</b>                    |
|                   | 40        | 47/50       | 414.03 (16.68)                          | 14/50      | 499.10 (11.21)                          | 0/50  | 0.00 (0.00)                             |
|                   | 80        | 50/50       | 1403.35 (11.67)                         | 0/50       | 0.00 (0.00)                             | 0/50  | 0.00 (0.00)                             |

# Default Hyperparameter Settings of S3-LRA-xNES

**Table 1: Default hyperparameter settings of S3-LRA-xNES as a function of  $d$ ,  $\eta_m^{\text{def}}$ ,  $\eta_\sigma^{\text{def}}$ , and  $\eta_B^{\text{def}}$ , where  $d$  is problem dimension and  $\eta_m^{\text{def}}$ ,  $\eta_\sigma^{\text{def}}$ , and  $\eta_B^{\text{def}}$  denote the default learning rates of xNES [Glasmachers 10].**

|                                                  |                                                                                                                    |                                                 |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| $\alpha_m = 2,$                                  | $\alpha_\sigma = 3 \left( \frac{d}{10} \right)^{\frac{\log(2.2)}{\log(8)}},$                                       | $\alpha_B = \min(2, 1 + \frac{16}{d}),$         |
| $\beta_m = 0.08,$                                | $\beta_\sigma^{\text{inc}} = 0.02, \beta_\sigma^{\text{dec}} = 0.1,$                                               | $\beta_B = 0.1,$                                |
| $\eta_m^{\text{min}} = 0.01\eta_m^{\text{def}},$ | $\eta_\sigma^{\text{min}} = 0.1 \left( \frac{d}{10} \right)^{\frac{\log(5.5)}{\log(8)}} \eta_\sigma^{\text{def}},$ | $\eta_B^{\text{min}} = 0.2\eta_B^{\text{def}},$ |
| $\eta_m^{\text{max}} = \eta_m^{\text{def}},$     | $\eta_\sigma^{\text{max}} = 1,$                                                                                    | $\eta_B^{\text{max}} = 1.5\eta_B^{\text{def}}.$ |
| $\eta_m^{\text{init}} = \eta_m^{\text{def}},$    | $\eta_\sigma^{\text{init}} = \eta_\sigma^{\text{def}},$                                                            | $\eta_B^{\text{init}} = \eta_B^{\text{def}},$   |
| $c_m = 0.01,$                                    | $c_\sigma = 0.01,$                                                                                                 | $c_B = 0.01,$                                   |

While these empirically derived settings yield strong performance, further investigation and theoretical analysis are required to enhance their robustness.